

Can dietary methionine restriction increase the effectiveness of chemotherapy in treatment of advanced cancer?

Most metastatic tumors, such as those originating in the prostate, lung, and gastrointestinal tract, respond poorly to conventional chemotherapy. Novel treatment strategies for advanced cancer are therefore desperately needed. Dietary restriction of the essential amino acid methionine offers promise as such a strategy, either alone or in combination with chemotherapy or other treatments. Numerous in vitro and animal studies demonstrate the effectiveness of dietary methionine restriction in inhibiting growth and eventually causing death of cancer cells. In contrast, normal host tissues are relatively resistant to methionine restriction. These preclinical observations led to a phase I clinical trial of dietary methionine restriction for adults with advanced cancer. Preliminary findings from this trial indicate that dietary methionine restriction is safe and feasible for the treatment of patients with advanced cancer. In addition, the trial has yielded some preliminary evidence of antitumor activity. One patient with hormone-independent prostate cancer experienced a 25% reduction in serum prostate-specific antigen (PSA) after 12 weeks on the diet, and a second patient with renal cell cancer experienced an objective radiographic response. The possibility that methionine restriction may act synergistically with other cancer treatments such as chemotherapy is being explored. Findings to date support further investigation of dietary methionine restriction as a novel treatment strategy for advanced cancer.